MAT2002 **Name (Print):** _____
Spring 2026
HW1
Deadline: 1st Feb

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1. Determine the order of $\frac{d^2y}{dt^2} + \sin(t+y) = \sin t$ also state whether the equation is linear or nonlinear.

2. Verify that $y = e^{t^2}\int_0^t e^{-s^2}ds + e^{t^2}$ is a solution of $y' - 2ty = 1$.

3. Solve the IVP: $y' - 2y = e^{2t}, \quad y(0) = 2.$

4. (Variation of Parameters) Consider the following method of solving the general linear equation of first order:
$$y' + p(t)y = g(t).$$
   (a) If $g(t) = 0$ for all t, show that the solution is
$$y = A\exp\left[-\int p(t)dt\right],$$
   where A is a constant.
   (b) If $g(t)$ is not everywhere zero, assume that the solution of Eq. (i) is of the form
$$y = A(t)\exp\left[-\int p(t)dt\right] \tag{1}$$
   where A is now a function of t. By substituting for y in the given differential equation, show that $A(t)$ must satisfy the condition
$$A'(t) = g(t)\exp\left[\int p(t)dt\right] \tag{2}$$
   (c) Find $A(t)$ from Eq. (2). Then substitute for $A(t)$ in Eq. (1) and determine y. Verify that the solution obtained in this manner agrees with that of $y = \frac{1}{\mu(t)}\left[\int_{t_0}^t \mu(s)g(s)ds + c\right]$.

5. Use the method above to solve the given differential equation $y' - 2y = t^2e^{2t}$.

6. Solve $\frac{dy}{dx} = \frac{x^2}{1+y^2}$.

7. Solve the equation
$$\frac{dy}{dx} = \frac{ay+b}{cy+d},$$
   where a, b, c, and d are constants.

8. Suppose that a tank containing a certain liquid has an outlet near the bottom. Let $h(t)$ be the height of the liquid surface above the outlet at time t. Torricelli's principle states that the outflow velocity v at the outlet is equal to the velocity of a particle falling freely (with no drag) from the height h.

(a) Show that $v = \sqrt{2gh}$, where g is the acceleration due to gravity.

(b) By equating the rate of outflow to the rate of change of liquid in the tank, show that $h(t)$ satisfies the equation

$$A(h)\frac{dh}{dt} = -\alpha a\sqrt{2gh},$$

where $A(h)$ is the area of the cross section of the tank at height h and a is the area of the outlet. The constant α is a contraction coefficient that accounts for the observed fact that the cross section of the (smooth) outflow stream is smaller than a. The value of α for water is about 0.6 .

(c) Consider a water tank in the form of a right circular cylinder that is 3 m high above the outlet. The radius of the tank is 1 m , and the radius of the circular outlet is 0.1 m . If the tank is initially full of water, determine how long it takes to drain the tank down to the level of the outlet.

9. Suppose that a rocket is launched straight up from the surface of the earth with initial velocity $v_0 = \sqrt{2gR}$, where R is the radius of the earth. Neglect air resistance.

(a) Find an expression for the velocity v in terms of the distance x from the surface of the earth.

(b) Find the time required for the rocket to go $240,000$mi (the approximate distance from the earth to the moon). Assume that $R = 4000$mi.

10. Solve $t^2y' + 2ty - y^3 = 0, \quad t > 0$

11. Solve $y' = ry - ky^2, r > 0$ and $k > 0$.

12. Determine whether each of the following equations is exact. If it is exact, find the solution

(a) $(x\ln y + xy) + (y\ln x + xy)y' = 0; \quad x > 0, \quad y > 0$

(b) $\frac{x}{(x^2+y^2)^{3/2}} + \frac{y}{(x^2+y^2)^{3/2}}\frac{dy}{dx} = 0$

13. Find the value of b for which the given equation is exact, and then solve it using that value of b.

(a) $\left(xy^2 + bx^2y\right) + (4x+y)x^2\frac{dy}{dx} = 0$

(b) $\left(ye^{2xy} + 5x\right) + bxe^{2xy}\frac{dy}{dx} = 0$